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CS415:Ethical Hacking

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UNIT-IV

Web Server Hacking: Web Applications and Database Attacks: Web Server Hacking, Web Application Hacking, Database Hacking

Wireless Technologies: Mobile Security and Attacks: Wireless Technologies, Mobile Device Operation and Security, Wireless LANs

IDS, Firewalls and Honeypots: Intrusion Detection Systems, Firewalls, Honeypots

Web Server Hacking

Web server hacking involves **exploiting vulnerabilities in the web server software, misconfigurations, and network to gain unauthorized access, control, or data.**

Common goals include **stealing data, deploying malware, or taking down a site.**

Web Server Concepts

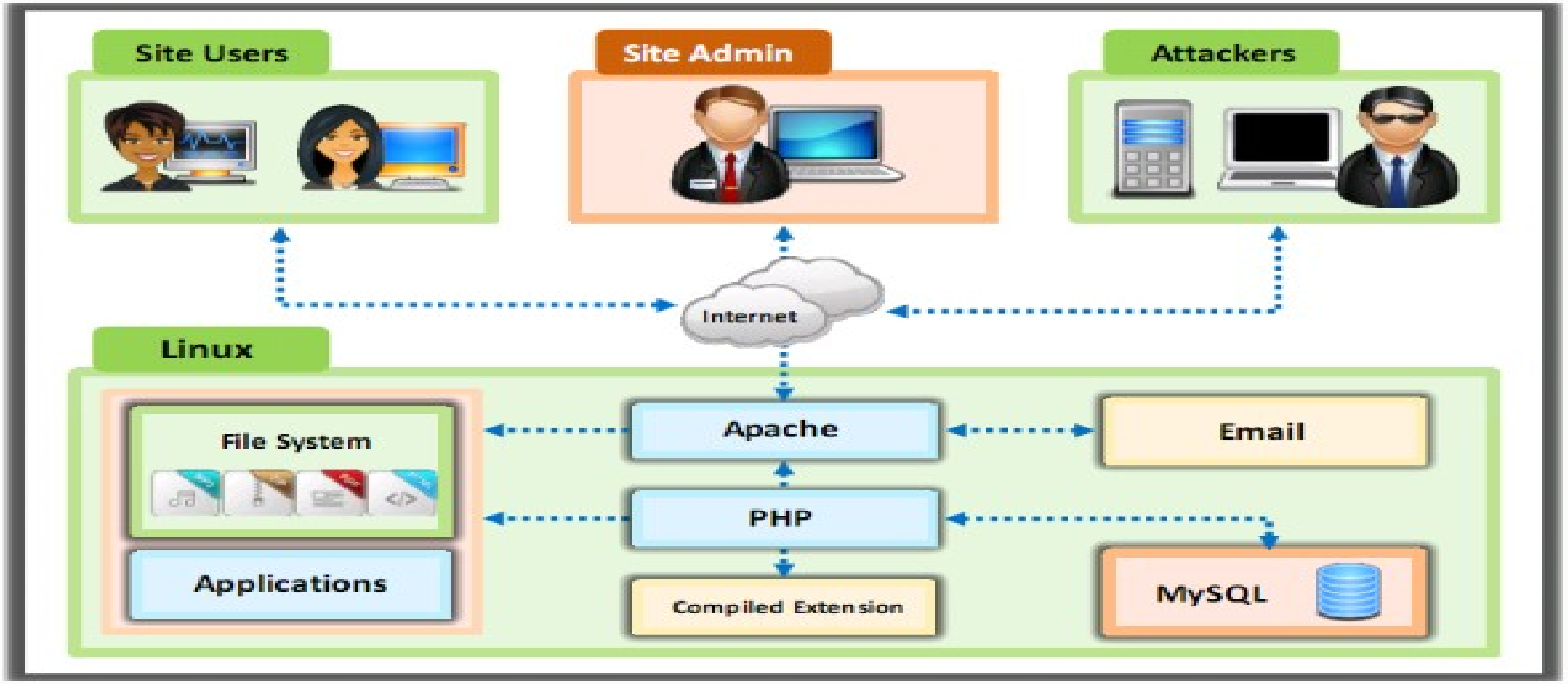
Web Server Operations

A web server is a computer System that stores processes and delivers web pages to clients via HTTP

Components of a Web Server

- **Document Root**
  - Store critical HTML files related to the web pages of a domain name that will be served in response to the request
- **Server Root**
  - Stores servers configuration error executable and log files
- **Virtual Document Tree**
  - Provides storage on a different machine or disk after the original disk is filled up
- **Virtual Hosting**
  - Technique of hosting multiple domains or websites on the same server
- **Web Proxy**
  - Proxy server that sits in between the web client and web server to prevent IP blocking and maintain anonymity

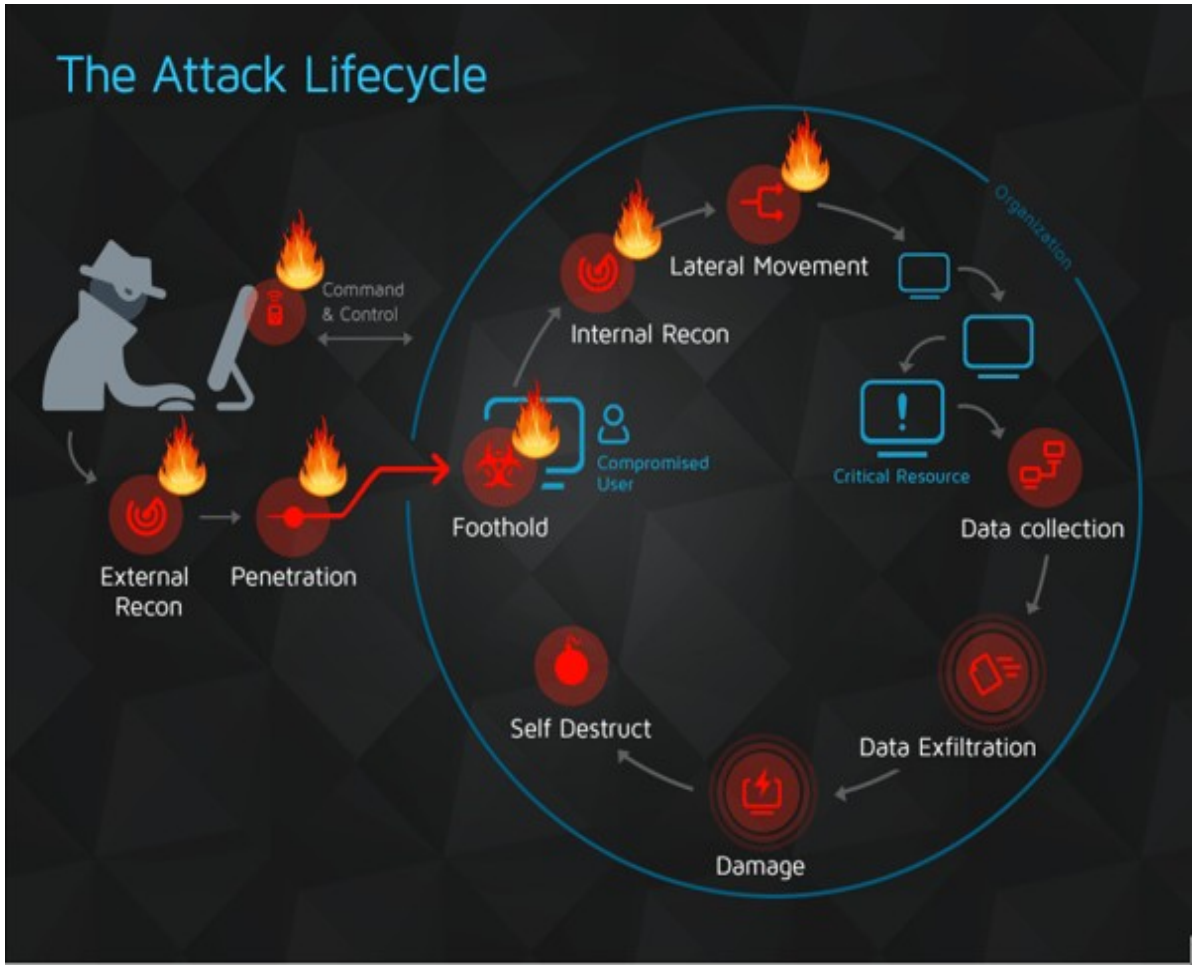
Open Source Web Server Architecture



Scanning Web Servers

This includes identifying live hosts, open ports, and services using tools like Nmap or Nessus.

Helps attackers map the server and discover exploitable entry points.



Banner Grabbing and Enumeration

Involves extracting metadata and information from services running on open ports.

Tools like Netcat or Telnet are used to gather server versions, OS details, and more.

Types of Banner Grabbing techniques

Active Banner Grabbing

An attacker sends the crafted packets to the victim's remote host and then waits for the response in the active banner grabbing technique.

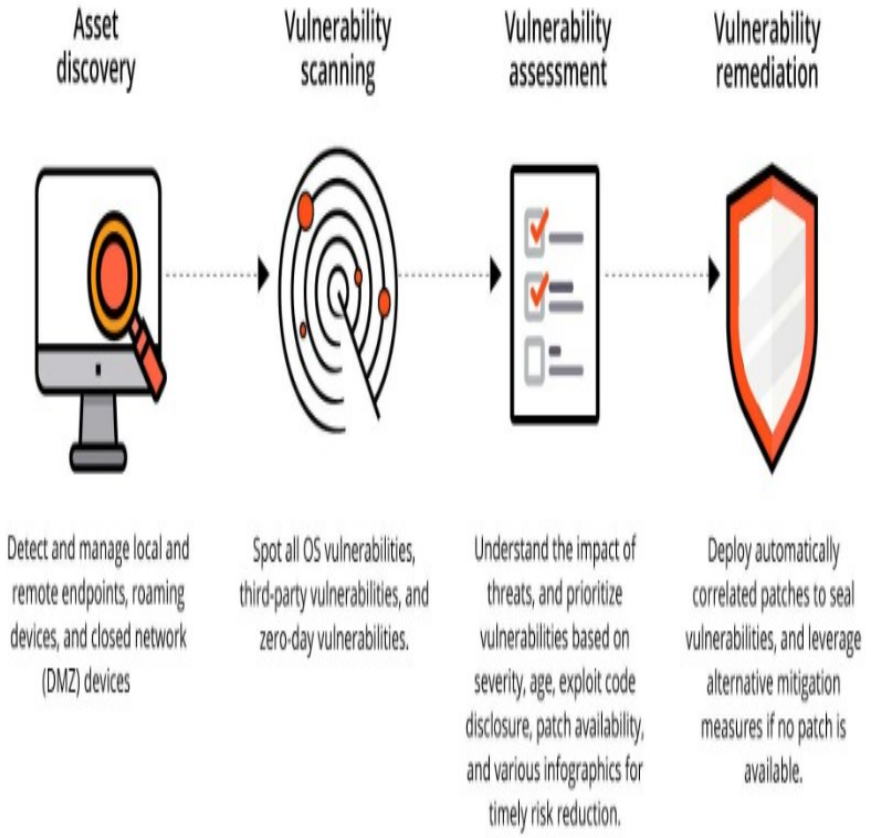
Passive Banner Grabbing

Passive banner grabbing relies on search engines, discovered banner data already available for use, remote service detection based on scans by online portal doors, and other components.

Web Server Vulnerability Identification

Detects known CVEs or misconfigurations in the web server software.

Vulnerability scanners and manual testing are common methods.



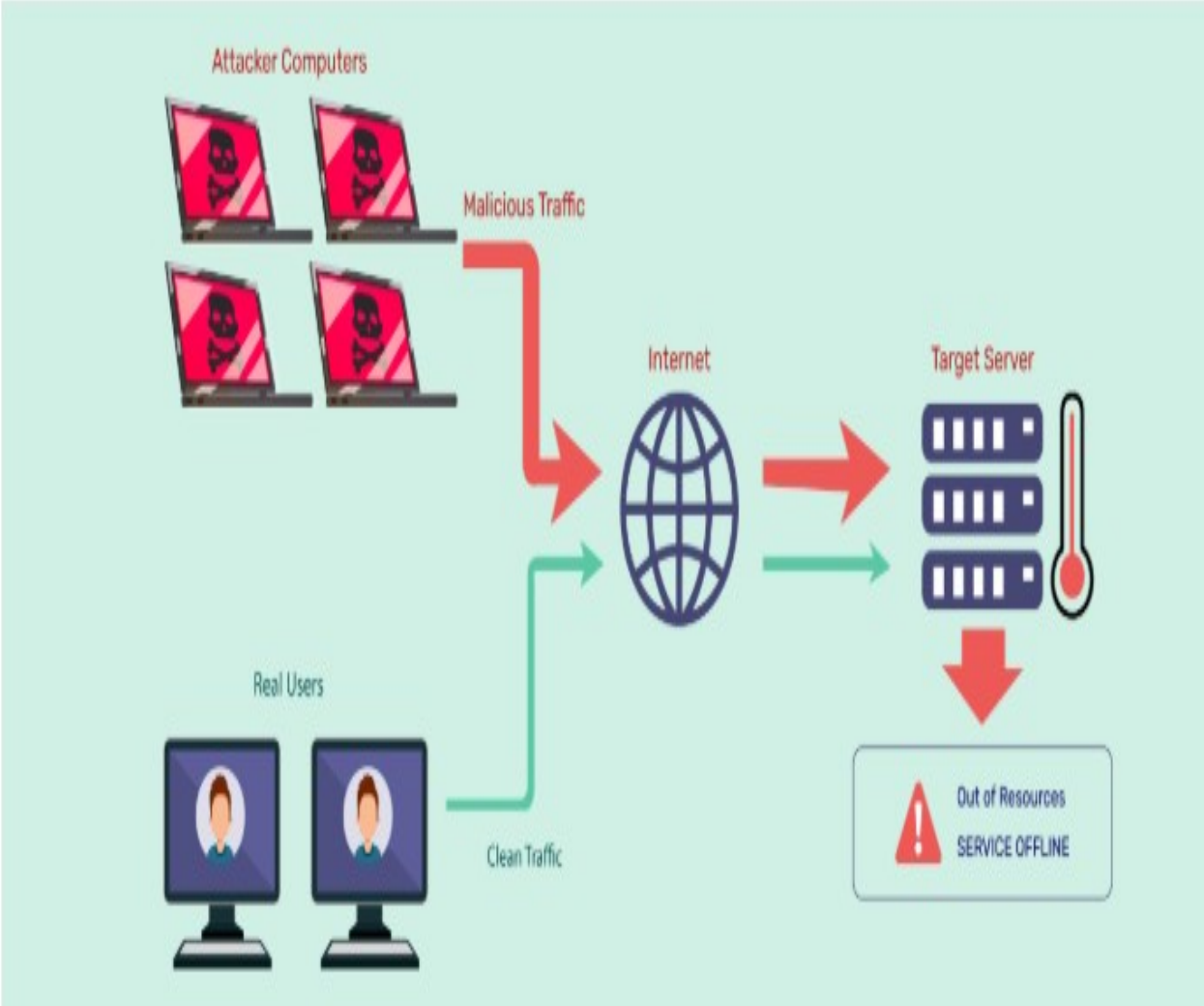
Attacking the Web Server

This can include brute force, buffer overflow, privilege escalation, or remote code execution aimed at compromising the server or its services.

DoS/DDoS Attack

Flooding the server with traffic to exhaust resources and disrupt services. DDoS uses multiple systems for amplification.

A DDoS attack works by overwhelming a target system, such as a website or server, with an excessive amount of traffic.



A DDoS attack typically involves the following steps:

1. Botnet Formation: The attacker assembles a network of compromised devices called a **botnet**. These devices, which can include computers, servers, IoT devices, or smartphones infected with malware, are under the attacker's control and used to launch the attack.

2. Reconnaissance: The attacker identifies potential targets to disrupt or disable, such as websites, servers, or network infrastructure. This can be done through automated scanning or by gathering information about vulnerable targets.

3. Command and Control (C&C) Setup: The attacker sets up a command-and-control infrastructure to communicate with the compromised devices in the botnet. This infrastructure allows the attacker to issue commands and coordinate the attack.

4. Initiation of the Attack: The attacker instructs the botnet's compromised devices to send massive amounts of traffic or requests to the target system. This flood of traffic overwhelms the target's resources, such as bandwidth, processing power, or memory.

5. Traffic Redirection: To amplify the attack, the attacker may use techniques like IP spoofing or reflection/amplification attacks. IP spoofing involves forging the source IP address of the attacking traffic to make it appear as if it's coming from legitimate sources.

Reflection/amplification attacks exploit vulnerable servers or services that respond with more traffic than they receive, magnifying the attack's scale.

A DDoS attack typically involves the following steps:

6. Impact on the Target: The target system's resources become overwhelmed due to the excessive traffic volume. As a result, the target system experiences performance degradation or becomes completely inaccessible to legitimate users. This disruption can lead to financial loss, damage to reputation, and potential security vulnerabilities.

7. Attack Persistence: The attacker may attempt to sustain the attack over an extended period, adjusting the attack strategy or targeting different parts of the infrastructure to evade DDoS detection or mitigation measures.

Types of DDoS Attacks

1. SYN Flood

A **SYN Flood** is a DDoS attack targeting the TCP (Transmission Control Protocol) handshake process. The attacker sends many TCP SYN packets to the target server but needs to complete the three-way handshake process. This causes the server to allocate resources for each incoming connection attempt but never frees them up, ultimately causing the server to become unresponsive.

2. LAND attack

A **LAND attack** is a type of DDoS attack that exploits a vulnerability in the TCP/IP protocol stack. The attacker sends a spoofed TCP SYN packet where both the source and destination IP addresses are set to the target server's IP address.

Additionally, random source and destination port numbers are used. This causes the target server to send a SYN-ACK packet back to itself, creating a loop that can render the server unresponsive.

3. ACK & PUSH ACK Flood

An ACK & PUSH ACK Flood is a DDoS attack targeting the TCP protocol by sending many TCP ACK or TCP PUSH packets to the target server. This can cause the server to become overwhelmed and unresponsive.

4. Fragmented ACK Flood

A **Fragmented ACK Flood** attack is a type of Distributed Denial of Service (DDoS) attack that targets the network and transport layers of the Internet Protocol (IP) suite. This attack exploits the way TCP/IP handles fragmented packets, specifically the acknowledgment (ACK) packets, to overwhelm and disrupt the target's network infrastructure.

Types of DDoS Attacks

5. Spoofed Session Flood

A **Spoofed Session Flood**, also known as a Fake Session Attack, overwhelms a target system by creating a large number of fake sessions using spoofed IP addresses. These spoofed requests appear to be from legitimate sources, causing the server to allocate resources to handle the influx of new session initiations. As the server tries to manage the flood of fake sessions, its resources, such as memory and CPU, become exhausted, leading to performance degradation or complete inaccessibility for legitimate users.

6. UDP Flood

A **UDP Flood** overwhelms a target system with a large volume of User Datagram Protocol (UDP) packets. This attack exploits the connectionless nature of UDP, which does not require a handshake process like TCP, making it easier to flood the target with unsolicited traffic.

7. DNS Flood

A **DNS Flood attack** targets Domain Name System (DNS) servers by overwhelming them with a high volume of DNS requests. Attackers send a massive number of requests to the DNS servers, often with spoofed IP addresses, causing the servers to become overwhelmed and unable to respond to legitimate queries. This results in disrupted services and inaccessibility of websites that rely on the targeted DNS servers for name resolution, leading to potential downtime and loss of business for affected websites.

8. VoIP Flood

A **VoIP Flood DDoS attack** targets Voice over Internet Protocol (VoIP) systems by overwhelming them with a large volume of SIP (Session Initiation Protocol) or RTP (Real-time Transport Protocol) packets. Excessive traffic disrupts normal call processing and depletes the system’s resources, causing significant degradation in call quality, dropped calls, or complete service unavailability.

Types of DDoS Attacks

9. CHARGEN Flood

A **CHARGEN Flood** exploits the Character Generator Protocol (CHARGEN), which is typically used for testing and debugging purpose. Attackers send spoofed requests to a server running CHARGEN, which then responds by generating a stream of characters based on the received request.

Attackers take advantage of CHARGEN responses being larger than the initial requests to increase the traffic volume aimed at the target server.

10. SSDP Flood

In **SSDP Flood**, attackers exploit SSDP's multicast capabilities by sending a high volume of spoofed SSDP packets to numerous devices on a network. This flood of packets overwhelms the devices' resources, causing network congestion and potentially leading to service degradation or unavailability.

11. SNMP Flood (SNMP Amplification)

An **SNMP Flood DDoS attack** overwhelms devices that run the Simple Network Management Protocol (SNMP) by flooding them with a barrage of SNMP requests. Attackers exploit SNMP's management capabilities to send a high volume of requests, exhausting the target device's processing capacity and network bandwidth.

12. HTTP Flood

An **HTTP Flood DDoS attack** overwhelms a web server with a huge number of requests. Attackers use botnets to send so many requests that the server can't handle them all, causing the website to slow down or crash. This disrupts normal access for users and can lead to financial losses and damage to the website's reputation. Defending against HTTP Floods involves using tools like firewalls and specialized services to filter out malicious traffic and keep the website accessible.

Types of DDoS Attacks

13. Recursive HTTP GET Flood

A Recursive HTTP GET Flood DDoS attack involves sending an overwhelming number of HTTP GET requests to a web server. Unlike a typical HTTP Flood where requests are straightforward, in a recursive attack, each request is crafted to trigger the server to generate subsequent requests to itself or to another server in a loop. This recursive behavior exponentially increases the load on the server, quickly consuming its resources such as bandwidth, CPU, and memory.

14. ICMP Flood

An **ICMP Flood DDoS attack** overwhelms a network or device with a large number of ping requests. Attackers send so many requests that the target's bandwidth and processing power get maxed out, causing slowdowns or making services unavailable. Defenses involve setting limits on how many ping requests can be processed and filtering out excessive traffic to keep networks running smoothly.

15. Smurf Attack

A **Smurf DDoS Attack** targets networks by exploiting the ICMP protocol's broadcast feature. The attacker sends a large amount of ICMP packets with a spoofed source IP address to a network's broadcast address, causing all devices on the network to respond to the request and flood the victim's network with traffic.

16. Fraggle Attack

A **Fraggle attack** is a type of DDoS attack that uses UDP packets to flood the victim's network with traffic. It is similar to a Smurf attack, but instead of using ICMP packets, it uses UDP packets to amplify the attack. The attacker sends a request to a vulnerable UDP service, such as the charge service, and the response is amplified and sent to the victim's IP address. This causes the victim's network to become overloaded and unresponsive.

Types of DDoS Attacks

17. Ping of Death Attack

A **Ping of Death** sends an oversized ICMP packet to the victim's network or web server. The packet size is larger than the maximum allowable size of an ICMP packet, causing the victim's system to crash or become unresponsive. This attack has existed for decades but is still effective against vulnerable systems.

18. Slowloris

A **Slowloris** attack is a type of DDoS attack that targets web servers by opening multiple connections and keeping them open for as long as possible. The attacker sends partial HTTP requests to the victim's web server but never completes them, keeping the connections open. This exhausts the server's resources and makes it unable to respond to legitimate requests.

19. ReDoS

A ReDoS attack, or Regular Expression Denial of Service, is a type of DDoS attack that targets vulnerable applications that use regular expressions to parse user input. The attacker sends malicious input to the application that triggers a regular expression with exponential matching possibilities, causing the application to enter an infinite loop and consume all available resources.

DNS Server Hijacking and DNS Amplification Attacks

Redirecting DNS queries or amplifying requests to overload servers or redirect traffic maliciously.

DNS hijacking and DNS amplification attacks are both forms of cyberattacks that exploit vulnerabilities in the Domain Name System (DNS) infrastructure. DNS hijacking redirects traffic to malicious websites, while DNS amplification uses open DNS resolvers to overwhelm a target with traffic.

DNS Hijacking:

DNS hijacking occurs when an attacker redirects a user's DNS queries to a malicious server, typically to steal data or redirect users to fraudulent websites.

How it works:

Attackers can achieve this by compromising local DNS settings on a user's computer, router, or by exploiting vulnerabilities in DNS servers.

Impact:

Redirects users to fake websites, potentially leading to data breaches, financial loss, and malware infections.

Example:

An attacker could modify a user's router's DNS settings to point to

DNS Amplification Attacks:

A type of DDoS attack where attackers use open DNS resolvers to send large volumes of DNS responses to a target, overwhelming their system.

How it works:

Attackers send DNS queries to open resolvers, spoofing the source IP address to be that of the intended victim. The large response from the DNS resolver is then sent to the victim, amplifying the attack traffic.

Impact:

Overloads the target's network, making it unavailable to legitimate users.

Example:

Key Differences:

Goal:

DNS hijacking aims to redirect traffic, while DNS amplification aims to overwhelm a target with traffic.

Target:

DNS hijacking targets individual users or networks, while DNS amplification targets the network infrastructure of a specific website or service.

Mechanism:

DNS hijacking involves redirecting traffic, while DNS amplification involves amplifying traffic using open DNS resolvers.

Prevention:

DNS Hijacking:

Use strong passwords on routers, keep firmware updated, educate users about phishing attacks, and monitor DNS settings.

DNS Amplification:

Limit recursion to authorized clients (internal network only), use DNSSEC, and monitor DNS traffic for unusual patterns.

Directory Traversal

Trick the server into revealing restricted directories and files using path manipulation (e.g., ../../etc/passwd).

Directory traversal, also known as **path traversal,** is a web security vulnerability that allows attackers to access files and directories outside the intended web server root. By manipulating URLs or input parameters, attackers can trick the server into revealing sensitive information, executing commands, or even gaining full control of the system.

Directory Traversal

Unauthorized Access:

Directory traversal attacks exploit flaws in how web applications handle file paths, enabling attackers to navigate beyond the designated web root directory.

Exploiting Input:

Attackers use special characters like ../ (or URL-encoded versions) to move up the directory tree and access files and directories they shouldn't be able to reach.

Vulnerable Applications:

Applications that incorporate user-supplied data into file paths without proper validation are susceptible to these attacks.

Directory Traversal – Risks:

Data Exposure:

Attackers can read sensitive files, including configuration files, databases, and application source code.

Code Execution:

In some cases, directory traversal can be combined with file upload vulnerabilities, allowing attackers to execute malicious code on the server.

System Compromise:

If attackers gain control of the web server, they can potentially take over the entire system.

Directory Traversal – Example:

Example:

Imagine a website where users can download files using URLs like:

https://example.com/download.php?file=report.pdf.

If the application doesn't properly sanitize the file parameter, an attacker could try **https://example.com/download.php?file=../../../../../etc/passwd** (on Linux) to access the /etc/passwd file, which contains user account information.

Prevention:

Input Validation: Thoroughly validate all user inputs that are used in file paths.

Whitelisting: Instead of blacklisting potentially harmful characters, create a whitelist of allowed characters or file paths.

Least Privilege: Run web applications with the least privileges necessary to minimize the potential damage from an attack.

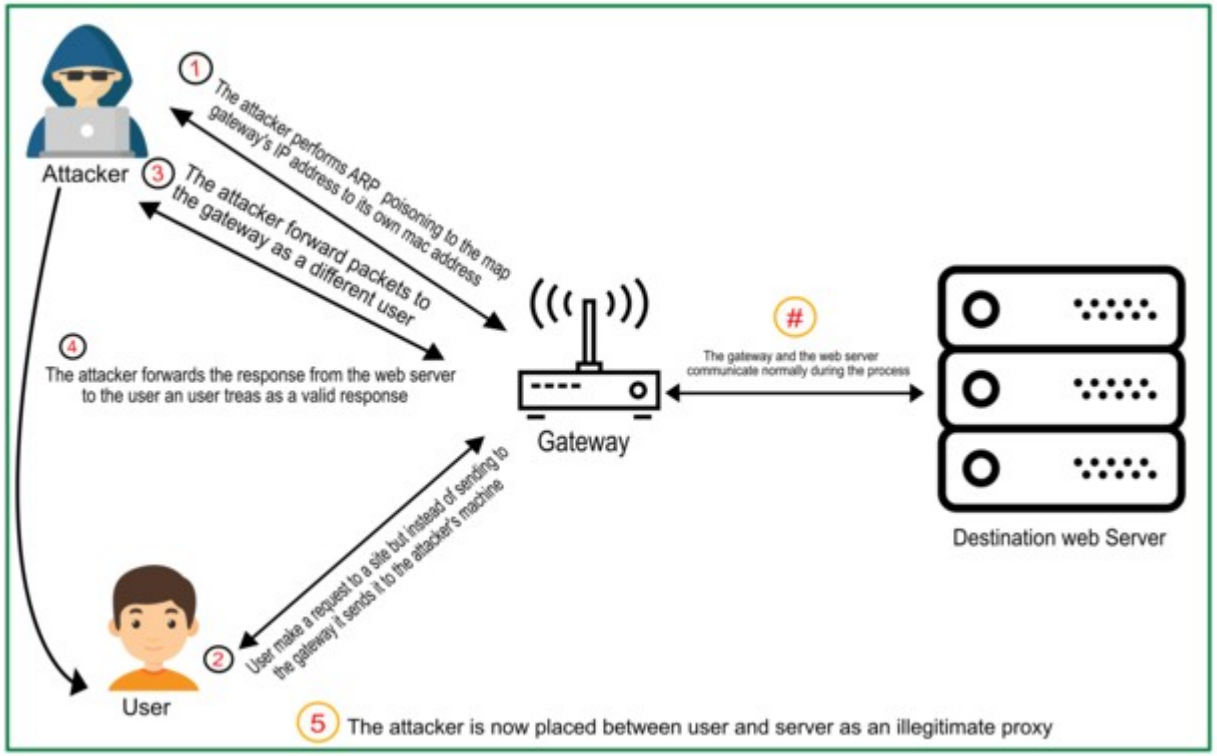
Regular Audits: Conduct regular security audits and penetration testing to identify and fix vulnerabilities.

Web Application Firewalls (WAFs): Deploy WAFs to filter malicious traffic and block directory traversal attempts.

Man-in-the-Middle Attacks

Intercepting data between the client and server to steal or manipulate data.

Often executed using ARP spoofing or rogue proxies.



Website Defacement

Altering the visual appearance of a website, typically by uploading unauthorized files or modifying site content.

Web Server Misconfiguration

Improper settings like default credentials, verbose error messages, or unused services can expose the server to attack.

HTTP Response Splitting

Injecting headers or malicious scripts into the response header to manipulate user experience or hijack sessions.

Web Server Password Cracking

Using brute-force or dictionary attacks against login interfaces or services like FTP or SSH to gain access.

IIS Vulnerabilities

Microsoft's Internet Information Services (IIS) has had critical CVEs like buffer overflows and misconfigurations attackers exploit.

Automated Exploit Tools

Tools like Metasploit, Nikto, or Burp Suite automate finding and exploiting server vulnerabilities.

Securing Web Servers

Best practices include patching, disabling unused services, enforcing strong authentication, and using firewalls and IDS.

Securing Web Servers

Keep your system up to date: Not updating the software regularly makes it weaker and leaves the system more vulnerable to attacks. Hackers take advantage of these flaws, and cybercriminals take advantage of them to get access to your network.

Prevent connecting to the public WiFi network: An unsecured Wi-Fi connection can be used by hackers to spread malware. If you allow file-sharing across a network, a hacker can simply infect your computer with tainted software. The ability of a hacker to put himself between you and the connection point poses the greatest threat to free Wi-Fi security.

Install Anti-virus, and update it regularly: Antivirus software is designed to identify, block, and respond to dangerous software, such as viruses, on your computer. Because computers are continuously threatened by new viruses, it is critical to keep antivirus software up to date.

Anti-virus updates include the most recent files required to combat new threats and safeguard your machine. These signature files are provided on a daily basis, if not more frequently.

Use IDS and firewall with updated signatures: NIDS are security threat detection and prevention systems that identify and prevent security threats from infiltrating secure networks. The use of NIDS has a negligible effect on network performance. NIDS are typically passive devices that listen to a network without interfering with the network's normal operation.

Backup your data: The fundamental purpose of a data backup is to keep a safe archive of your vital information, whether it's classified documents for your business or priceless family photos so that you can quickly and effortlessly recover your device in the event of data loss. Backup copies allow data to be restored from a previous point in time, which can aid in the recovery of a business after an unanticipated occurrence. Protecting against primary data loss or corruption requires storing a copy of the data on a secondary medium.

Install a Firewall: Firewalls defend your computer or network from outside cyber attackers by filtering out dangerous or superfluous network traffic. Firewalls can also prevent harmful malware from gaining internet access to a machine or network.

Web Application Hacking

Targeting the app layer with attacks such as injections, session hijacking, and exploiting weak logic in the application.

Unvalidated Input

Allowing unfiltered user input into backend systems opens doors to XSS, SQL injection, and command injection.

Parameter/Form Tampering

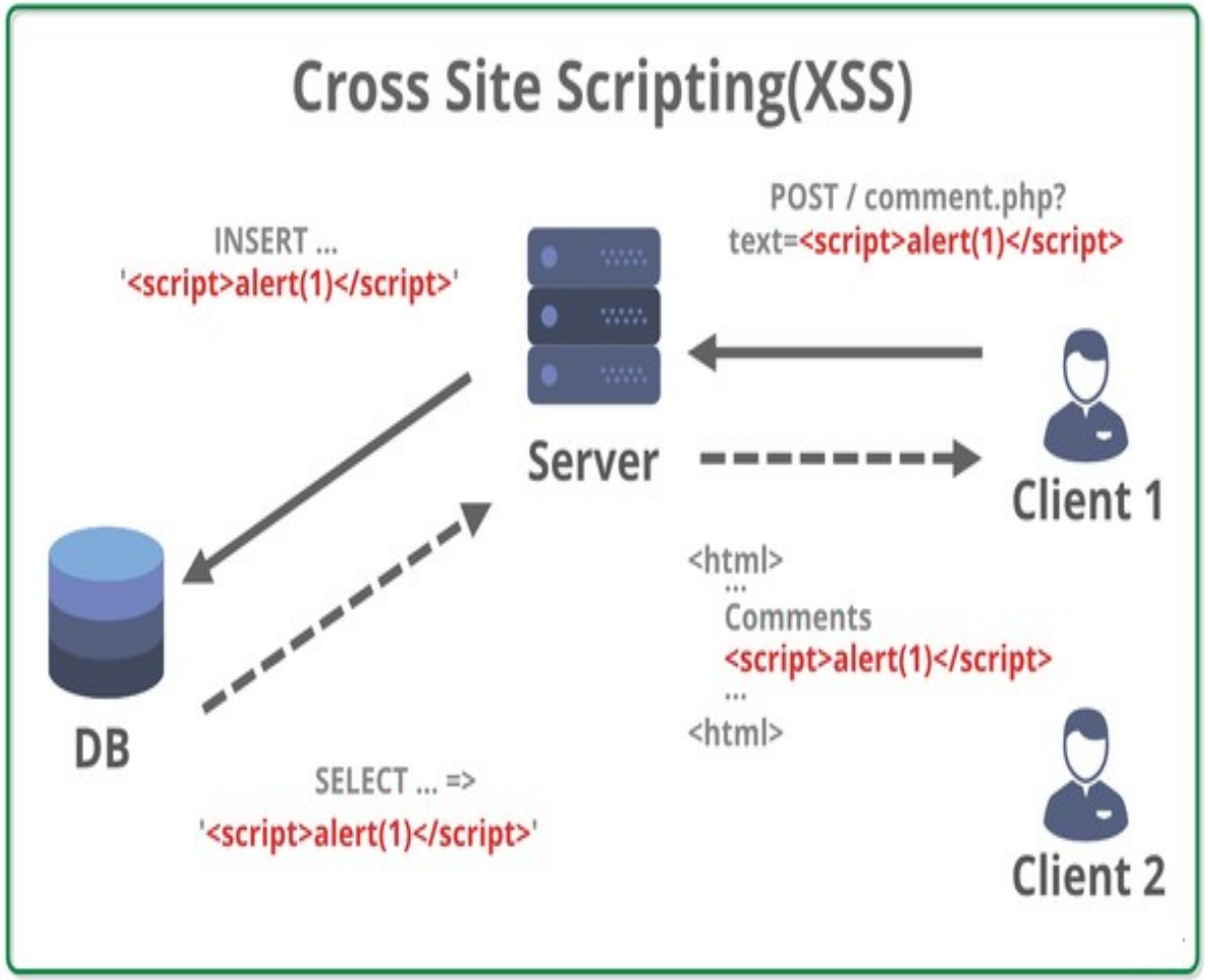
Manipulating URL parameters or form data to bypass controls or elevate privileges.

Injection Flaws

Injecting malicious code (SQL, OS, LDAP) into input fields to trick the server into executing unintended commands.

Cross-Site Scripting and Cross-Site Request Forg

XSS allows attackers to inject scripts into web pages. CSRF tricks users into submitting malicious requests unknowingly.



Hidden Field Attacks

Modifying hidden form fields to manipulate backend processes like prices, roles, or permissions.

Other Web Application Attacks

Include logic flaws, broken access controls, session fixation, insecure file uploads, and more.

Attacking Web-Based Authentication

Involves brute force, credential stuffing, session hijacking, or exploiting weak authentication flows.

Web-Based Password Cracking and Authentication Attacks

Using automated tools to guess credentials or exploit authentication vulnerabilities like weak password resets.

Cookies

Can be manipulated or stolen via XSS to hijack sessions. Secure flags, encryption, and short lifetimes are essential.

URL Obfuscation

Using hexadecimal, octal, or base64 encoding to disguise URLs or bypass filters and security controls.

Intercepting Web Traffic

Using proxies like Burp Suite or Wireshark to analyze and manipulate HTTP/S traffic between the client and server.

Securing Web Applications

Use of input validation, secure coding practices, WAFs, regular testing, and adhering to OWASP guidelines.

Database Hacking

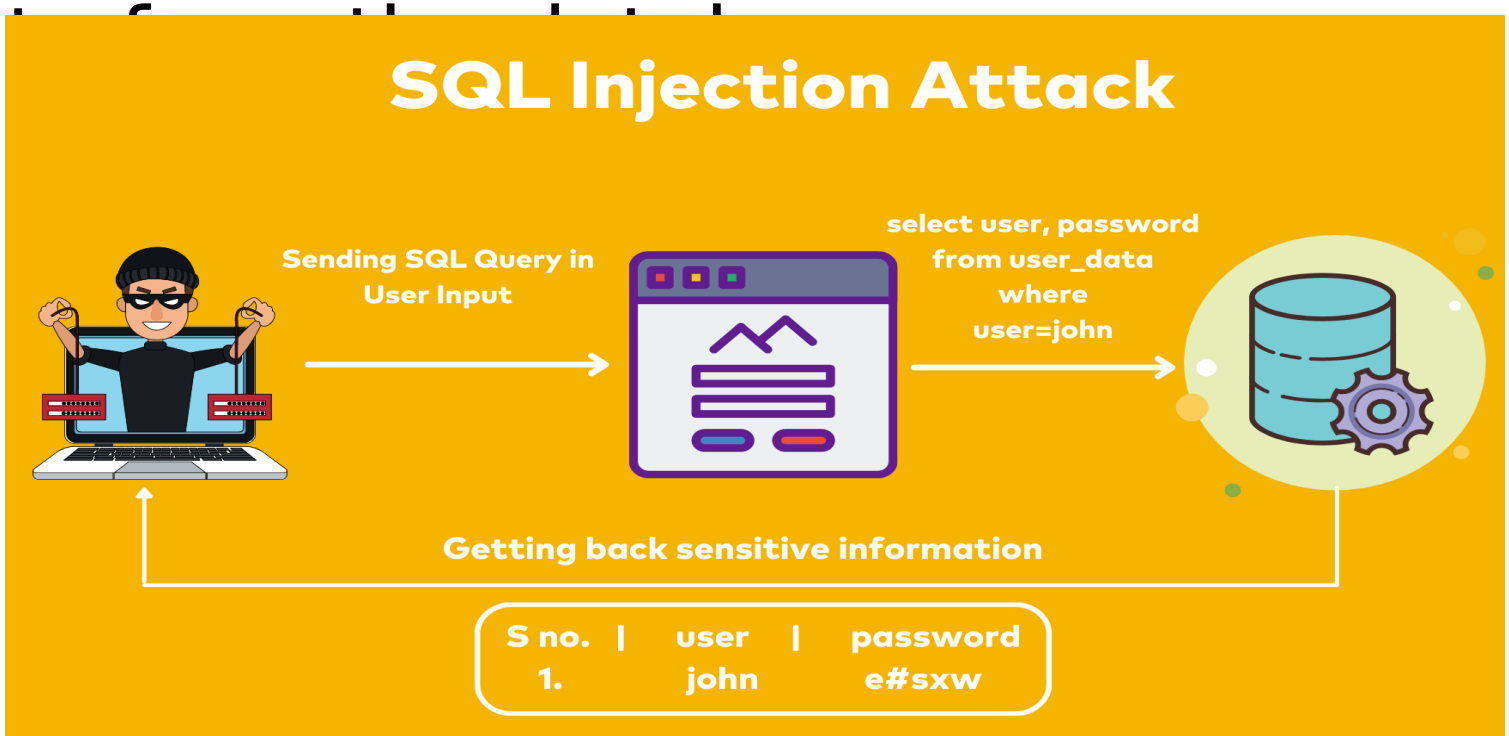
Attacking backend databases by exploiting misconfigurations, weak credentials, or injection vulnerabilities.

Identifying SQL Servers

Scanning and fingerprinting database servers to identify vendor, version, and accessible interfaces.

SQL Injection Vulnerabilities

Injecting SQL queries via user inputs to view, modify, or delete da



SQL Injection Hacking Tools

Tools like SQLmap, Havij, and jSQL automate SQL injection exploitation and data extraction.

Wireless Technologies: Mobile Security and Attacks: Wireless Technologies, Mobile Device Operation and Security, Wireless LANs

Mobile security is the practice of protecting mobile devices, and the data they contain, from a wide range of cyber threats that exploit vulnerabilities in wireless technologies, mobile operating systems, and user behavior.

As wireless networks and mobile devices have become essential for both personal and business use, the need for comprehensive security has become critical.

Mobile security and attacks involve protecting mobile devices and wireless networks, which are vulnerable to threats like eavesdropping on wireless signals, malware, and unauthorized access to local or network resources.

Wireless technologies like Wireless LANs (WLANs) can be attacked through their broadcast nature, and mobile devices are targeted by threats such as malicious apps, spyware, and social engineering. Security measures include device-level protection (like passwords and encryption) and network-level security to prevent attacks.

Wireless technologies and security

Nature of wireless: Wireless communication is broadcast, making it susceptible to eavesdropping and attacks by anyone within range.

Wireless LAN (WLAN): A local area network that uses wireless communication to link devices, such as in a home or office.

Security protocols: Encryption protocols like WEP were designed to secure WLANs by encrypting data, though older versions like WEP have significant vulnerabilities.

Attack surface: The ease of access provided by wireless networks creates a larger attack surface compared to

Attacks leveraging wireless technologies

The inherent broadcast nature of wireless signals makes them vulnerable to various forms of interception and interference.

Eavesdropping (Packet Sniffing): Attackers use packet sniffers to passively intercept and analyze data transmitted over unencrypted or poorly encrypted wireless networks. This can expose sensitive data such as login credentials, financial information, and personal communications.

Man-in-the-Middle (MitM) Attacks: In an MitM attack, an attacker places themselves between a user's mobile device and a legitimate access point. They can then intercept and alter communications without the user's knowledge. Mobile devices are

Rogue Access Points (Evil Twin): Attackers set up fake Wi-Fi access points that mimic legitimate ones, such as those in a cafe or airport. When a user's device automatically connects to this "evil twin" network, the attacker can intercept all traffic and potentially inject malware.

Wireless Jamming (Denial-of-Service): A denial-of-service (DoS) attack on a wireless network involves flooding the frequency spectrum with interference signals to prevent legitimate devices from communicating. Attackers can also use deauthentication attacks, which force devices to disconnect from a Wi-Fi network.

Bluejacking and Bluesnarfing: These attacks exploit vulnerabilities in Bluetooth connections. Bluejacking is the sending of unsolicited messages to a nearby device, while Bluesnarfing is

Mobile device operation and security

Definition: Mobile security is cybersecurity for devices like smartphones, tablets, and laptops, protecting them from threats like data loss, malware, and credential theft.

Device operation: Mobile devices are used for more than communication, acting as handheld computers that require robust security for professional and personal use.

Security measures: Basic security features include passwords, fingerprint sensors, and facial recognition. More advanced measures involve endpoint security, which monitors devices for malicious behavior.

Types of threats:

- Malicious applications and websites
- Data leaks
- Spyware

Mobile device operation and security

Mobile devices, such as smartphones and tablets, are complex computing platforms with their own unique security risks due to their constant connectivity, app ecosystems, and portability.

Malicious Applications (Malware): Hackers upload malicious apps to third-party app stores or use phishing to trick users into downloading them. These apps can steal personal information, install spyware, or create backdoors for remote control.

Phishing and Smishing: Phishing attacks target mobile users through deceptive emails, while smishing uses text messages. The smaller screens on mobile devices can hide malicious indicators, making users more likely to click on a fraudulent link.

Ransomware: This type of malware encrypts a user's data or locks their device, demanding a ransom for its release. Mobile ransomware has become more frequent with the increased use of devices for business.

Jailbreaking and Rooting: These techniques give users administrative access to the mobile operating system, bypassing built-in security protections. This practice removes security safeguards put in place by manufacturers and can make the device vulnerable to malware and exploits.

Wireless LANs and attacks

Vulnerability: WLANs are a significant target for attacks because of their open and accessible nature.

Consequences: Attacks can lead to data theft, unauthorized access, and service disruption.

Proximity is key: An attacker does not need to physically connect to the network but merely be within range to launch an attack.

Lack of awareness: A lack of knowledge about wireless network security contributes to increased vulnerability.

Wireless LANs

Wireless Local Area Networks (WLANs) provide flexible connectivity but introduce distinct security challenges compared to wired networks.

Outdated Encryption Protocols: Older wireless security standards, such as Wired Equivalent Privacy (WEP), contain well-known flaws that make them easy for attackers to crack. Even Wi-Fi Protected Access II (WPA2) has had vulnerabilities, though it remains a stronger option than WEP.

Weak Passwords and Default Configurations: Many Wi-Fi routers ship with default admin credentials and SSIDs that are easy for attackers to guess. If users don't change these settings, their network is at risk of unauthorized access.

Poorly Secured Guest Networks: Misconfigured or unsecured guest networks can provide a backdoor into a company's sensitive network.

Ad-Hoc Networks: These peer-to-peer networks bypass traditional access points and can create high security risks. Users may connect to an ad-hoc network without prior authentication, allowing unauthorized access to other

Best practices for mobile and wireless security

Wireless Local Area Networks (WLANs) provide flexible connectivity but introduce distinct security challenges compared to wired networks.

Strengthen authentication: Enable a strong, complex passcode or biometric authentication (like fingerprint or facial recognition) on all devices. Use multi-factor authentication (MFA) for apps and accounts whenever possible.

Keep software updated: Regularly apply all operating system and application updates. These often include critical security patches for newly discovered vulnerabilities.

Be cautious with public Wi-Fi: Avoid conducting sensitive transactions on unsecured public Wi-Fi networks. Use a Virtual Private Network (VPN) to encrypt traffic and protect your data when on public networks.

Best practices for mobile and wireless security

Manage app permissions: Restrict application permissions to only what is necessary for an app's function. Regularly review and revoke any unnecessary access.

Disable unused wireless features: Turn off Bluetooth, Wi-Fi, and location services when not in use to reduce the attack surface.

Download from trusted sources: Only download apps from official sources like the Apple App Store and Google Play Store, and always check reviews. Avoid unofficial app marketplaces.

Secure the WLAN: Change the default router name (SSID) and administrator password immediately. Use the strongest encryption protocol available, such as WPA3.

Back up your data: Enable regular backups of your mobile device data to ensure that important information can be restored if the device is lost, stolen, or compromised by ransomware.

Protect lost or stolen devices: Enable remote wipe and lock features to protect data in case a device is lost or stolen.

Intrusion Detection Systems, Firewalls, Honeypots

Firewalls, Intrusion Detection Systems (IDS), and honeypots are all cybersecurity tools that protect networks, but they function differently: firewalls are the first line of defense, blocking unwanted traffic based on rules, while IDS monitor network traffic for suspicious activity and send alerts.

Honeypots are decoy systems designed to lure attackers away from real assets and gather intelligence on their methods without risk to the actual network

Firewalls

Function:

A security barrier that controls incoming and outgoing network traffic based on a set of defined rules.

How it works:

It can allow or block traffic based on IP address, port, protocol, and other criteria.

Primary role:

To prevent unauthorized access by filtering and blocking malicious or non-legitimate traffic before it can reach the network.

Intrusion Detection Systems (IDS)

Function:

A monitoring system that analyzes network traffic for signs of malicious activity or policy violations.

How it works:

It uses methods like signature-based or anomaly-based detection to identify threats by comparing traffic to known attack patterns or flagging unusual behavior.

Primary role:

To alert administrators to threats that have bypassed other defenses, providing an extra layer of protection.

Honeypots

Function:

A decoy system or server that is intentionally made to look vulnerable to lure and trap attackers.

How it works:

It mimics legitimate systems to attract those trying to exploit vulnerabilities. When an attacker interacts with it, the honeypot logs the activity and alerts administrators.

Primary role:

To divert attackers from real assets, gather intelligence on their tools and techniques, and reduce the risk to critical data and systems.

Feature	Intrusion Detection System (IDS)	Firewall	Honeypot
Function	Detects and alerts. It monitors network traffic for malicious activity or policy violations and notifies administrators.	Filters and blocks. It controls incoming and outgoing network traffic based on a predefined set of security rules.	Deceives and gathers intelligence. It is a decoy system designed to attract and trap attackers away from real targets.
Nature	Passive. It does not prevent an attack itself but alerts you that one is happening or has occurred. An Intrusion Prevention System (IPS) is the active version that can block threats.	Active. It actively enforces policy by allowing or denying traffic.	Active deception. It engages attackers and monitors their behavior.
Scope	Monitors traffic that has already passed through the firewall, identifying threats that may have slipped past it.	Creates a barrier at the network perimeter, acting as a first line of defense to block unauthorized access.	Designed to observe and analyze threats that interact with the decoy system, often placed behind a firewall for added security.
Detection Method	Uses signature-based detection (known attack patterns) and anomaly-based detection (deviations from normal behavior).	Uses rules based on source/destination IP addresses, port numbers, and protocols.	Relies on the fact that any interaction is suspicious since the system has no legitimate users.
Primary Goal	To identify potential threats that have already breached or are attempting to breach the network and trigger an investigation.	To prevent unauthorized traffic from entering or leaving the network.	To divert attackers, waste their time, and gather intelligence on their methods to improve overall security.
Advantage	Adds a layer of security to catch threats that firewalls miss. Can detect internal threats or new, unknown attacks with anomaly-based detection.	Acts as a fundamental, first-line security barrier against many common threats.	Provides insight into new threats and attacker tactics in a safe, controlled environment.
	Cannot prevent attacks on its own. It can also produce false positives or false	Cannot stop attacks originating from inside the network.	Cannot protect against breaches on legitimate systems and may fail if an

How they work together

For comprehensive cybersecurity, these tools are most effective when used in combination.

A firewall acts as the first line of defense, filtering out a large portion of unwanted traffic.

An IDS, placed behind the firewall, monitors all traffic that passes through, watching for anything suspicious the firewall may have missed.

Honeypots can be strategically deployed within the network as decoys. If an attacker bypasses the firewall and IDS and engages with a honeypot, the security team is alerted and can analyze the